VISUAL JIGSAW POST-TRAINING IMPROVES MLLMS

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ABSTRACT

Reinforcement learning with verifiable rewards (RLVR) has emerged as a powerful post-training paradigm for Multimodal Large Language Models (MLLMs), yet existing approaches remain predominantly text-centric—using visual inputs merely as sparse contextual evidence for chain-of-thought reasoning. This skews optimization away from intrinsic visual perception, leaving fine-grained, temporal, and 3D spatial understanding under-optimized. We introduce **Visual Jigsaw** (VJ), a self-supervised, vision-centric post-training framework that recasts visual understanding as a lightweight, verifiable ordering task: visual elements are partitioned, shuffled, and the model must recover the correct permutation expressed as a text sequence. This formulation is naturally verifiable, requires no generative modules, and is instantiated across three modalities: (i) **Image Jigsaw**—recovering the spatial arrangement of 3×3 image patches on COCO; (ii) **Video Jigsaw**—ordering temporally shuffled video clips from LLaVA-Video; and (iii) **3D Jigsaw**—sorting depth-ordered RGB-D point sets from ScanNet. Training uses Group Relative Policy Optimization (GRPO) with a graded reward combining exact-match accuracy, discounted partial correctness, and strict format constraints. Extensive experiments on Qwen2.5-VL-7B show consistent gains across all three modalities—up to +6.06 on fine-grained image perception (MMStar), +5.23 on temporal video understanding (AoTBench), and +17.11 on depth-ordering (DA-2K)—while outperforming supervised fine-tuning and preserving reasoning capabilities in thinking models such as ThinkLite-VL. Visual Jigsaw provides a simple, scalable, and modality-general recipe for strengthening intrinsic visual understanding in MLLMs.

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1 INTRODUCTION

The recent surge of reinforcement learning with verifiable rewards (RLVR) has substantially advanced the reasoning capabilities of large language models (Guo et al., 2025; Shao et al., 2024). Inspired by these breakthroughs, researchers have extended RLVR to Multimodal Large Language Models (MLLMs) (Huang et al., 2025; Meng et al., 2025; Liu et al., 2025b; Feng et al., 2025), achieving notable gains on tasks involving mathematical reasoning, grounding, and visual question answering. Yet a critical gap persists: nearly all existing vision-language RLVR methods use images or videos primarily as sparse contextual evidence to support text-centric chains of thought. Visual perception itself—the fine-grained, temporal, and 3D spatial understanding that distinguishes powerful multimodal systems—is rarely the direct target of optimization.

This asymmetry leads to a fundamental under-optimization of core visual abilities. Models trained with reasoning-oriented RL may master logical deduction over visual cues while remaining weak at accurately perceiving local image details, understanding temporal order in videos, or reasoning about depth relationships in 3D scenes. Worse, recent work has highlighted that supervised fine-tuning (SFT) on perception tasks can actually degrade performance on other visual benchmarks (Chen et al., 2025), motivating the search for more principled, RL-based perception training signals.

Our key insight is that visual understanding can be recast as a *lightweight, verifiable ordering task*: given a set of shuffled visual elements, predict the correct permutation expressed as a comma-separated list of indices. This formulation—inspired by classical jigsaw puzzle pretraining (Noroozi & Favaro, 2016)—has three attractive properties: (1) the ground-truth permutation is instantly verifiable without any external judge, enabling direct use of RLVR; (2) solving the task requires genuine

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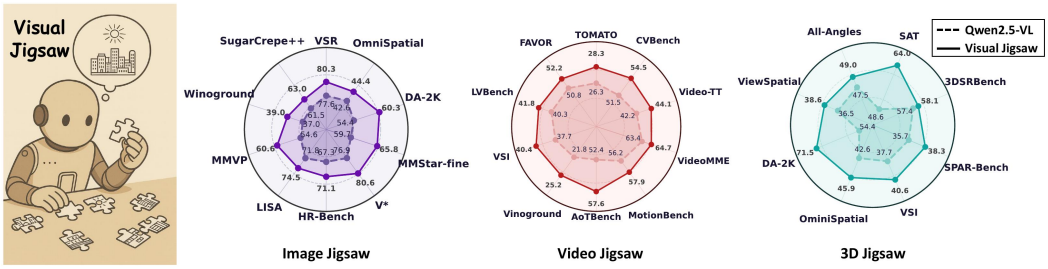

Figure 1: **Visual Jigsaw** is a self-supervised post-training task that enhances visual perception and understanding in MLLMs. By shuffling visual elements (image patches, video clips, or depth-ordered point sets) and training the model to recover the correct permutation, Visual Jigsaw provides a vision-centric, verifiable training signal compatible with RLVR without architectural changes.

visual understanding of local patterns, temporal dynamics, or depth ordering rather than surface-level language cues; and (3) the model outputs only text, making the approach plug-and-play for any text-output MLLM without generative modules or architectural modifications.

We propose **Visual Jigsaw** (VJ), a modality-general post-training framework that instantiates this ordering task across images, videos, and 3D data (see Figure 1 and Figure 2):

- **Image Jigsaw**: A 3×3 grid of patches from COCO (Lin et al., 2014) images is shuffled; the model predicts the correct spatial permutation.
- **Video Jigsaw**: Temporal clips sampled from LLaVA-Video (Zhang et al., 2024b) are re-ordered; the model infers the chronological sequence.
- **3D Jigsaw**: RGB-D frames from ScanNet (Dai et al., 2017) are sorted by depth; the model recovers the front-to-back ordering.

All three variants are trained with Group Relative Policy Optimization (GRPO) (Shao et al., 2024) using a carefully designed graded reward that combines exact-match accuracy, discounted partial correctness ($\gamma = 0.2$), a format constraint, and a repetition penalty to prevent reward hacking.

We evaluate Visual Jigsaw on Qwen2.5-VL-7B-Instruct (Bai et al., 2025) across a comprehensive suite of benchmarks. Image Jigsaw improves MMVP (Tong et al., 2024) by +6.00 and MMStar fine-grained (Chen et al., 2024) by +6.06. Video Jigsaw (64 frames) improves AoTBench (Xue et al., 2025) by +5.23 and Vinoground (Zhang et al., 2024a) by +3.40. 3D Jigsaw yields particularly striking gains on depth-ordering tasks: DA-2K (Yang et al., 2024) by +17.11 and SAT-Real (Ray et al., 2024) by +15.34. Crucially, RL consistently outperforms SFT, and applying Image Jigsaw to a reasoning-oriented model (ThinkLite-VL (Wang et al., 2025e)) enhances visual perception while preserving mathematical reasoning ability.

Our main contributions are:

1. We introduce Visual Jigsaw, a verifiable, self-supervised, vision-centric RL post-training task expressed entirely through text outputs, requiring no generative components or architectural changes.
2. We provide three concrete instantiations—image, video, and 3D—each targeting a different dimension of visual understanding (fine-grained perception, temporal ordering, and depth reasoning).
3. We design a graded reward function combining exact-match, partial correctness, format checking, and repetition penalties that stabilizes training and prevents reward hacking.
4. We demonstrate that Visual Jigsaw consistently outperforms SFT, generalizes broadly across perception benchmarks, and complements reasoning-centric RL without degrading existing reasoning skills.

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2 Related Work

2.1 Reinforcement Learning for MLLMs

RLHF (Ouyang et al., 2022) and RLVR have become standard paradigms for aligning and improving language models. DeepSeek-R1 (Guo et al., 2025) demonstrated that RLVR with verifiable rewards can elicit strong chain-of-thought reasoning. This inspired a wave of multimodal extensions: Vision-R1 (Huang et al., 2025) and MM-Eureka (Meng et al., 2025) apply rule-based RL to multimodal math benchmarks; Visual-RFT (Liu et al., 2025b) designs verifiable perception rewards for object detection and grounding; Seg-Zero (Liu et al., 2025a) tackles reasoning segmentation via RL; and Video-R1 (Feng et al., 2025) extends RLVR to video question answering. Concurrent works such as VL-Cogito (Yuan et al., 2025a) and LLaVA-Critic-R1 (Wang et al., 2025d) further scale reasoning-centric post-training.

Despite these advances, all of the above methods target language-level reasoning objectives with visual inputs playing a supporting role. In contrast, Visual Jigsaw makes visual perception itself the primary learning target, using permutation prediction as a verifiable, vision-centric training signal.

2.2 Self-Supervised Visual Representation Learning

Classical self-supervised learning methods have long leveraged proxy tasks to learn visual representations without manual labels. These include contrastive objectives (He et al., 2020; Chen et al., 2020), masked image modeling (He et al., 2022; Bao et al., 2021; Zhou et al., 2021), and predictive tasks such as colorization (Zhang et al., 2016), context prediction (Doersch et al., 2015), and rotation prediction (Gidaris et al., 2018). Jigsaw puzzle solving was introduced by Noroozi & Favaro (2016) as a pretraining proxy, with subsequent extensions to domain generalization (Carlucci et al., 2019) and video understanding (Huo et al., 2021; Wang et al., 2022). However, these approaches operate at the vision encoder level and are not designed for the autoregressive, text-output setting of modern MLLMs.

More recently, visual pretraining objectives have been integrated into MLLM training. ROSS (Wang et al., 2025b) and ASVR (Wang et al., 2025a) reconstruct visual representations during instruction tuning, while Pixel Reasoner (Su et al., 2025) and DeepEyes (Zheng et al., 2025) explicitly encourage pixel-level reasoning. Our work is complementary: we target the post-training stage and use RL rather than supervised reconstruction, avoiding the mode collapse and forgetting risks of SFT.

2.3 Jigsaw Puzzles for Visual Understanding

The jigsaw puzzle framework for visual pretraining (Noroozi & Favaro, 2016) has been extended in several ways: iterative jigsaw solving for harder puzzles (Wei et al., 2019), fine-grained patch discrimination (Du et al., 2020), and video-level temporal ordering (Ahsan et al., 2019; Huo et al., 2021). JigsaWR1 (Wang et al., 2025f) and concurrent works (Chen et al., 2023) explore jigsaw-style tasks for representation learning, while Scene-R1 (Yuan et al., 2025b) applies spatially grounded RL to 3D understanding. Visual Jigsaw unifies these threads into a single post-training recipe for modern instruction-tuned MLLMs under the RLVR paradigm.

3 Method

3.1 Problem Formulation

Let $\mathcal{V} = \{v_1, v_2, \ldots, v_n\}$ be a set of n visual elements (image patches, video clips, or depth-ranked frames) extracted from a raw visual input. We apply a random permutation $\sigma \in S_n$ to obtain the shuffled sequence $\mathcal{V}^\sigma = \{v_{\sigma(1)}, v_{\sigma(2)}, \ldots, v_{\sigma(n)}\}$. The Visual Jigsaw task asks the model to predict the inverse permutation σ^{-1}, expressed as a comma-separated list of indices indicating which position each shuffled element originally came from.

Formally, given the shuffled visual input $\mathcal{V}^\sigma$ and a natural language instruction, the model must output a permutation prediction $\hat{\pi} = [\hat{\pi}_1, \hat{\pi}_2, \ldots, \hat{\pi}_n]$ wrapped in a reasoning chain followed by

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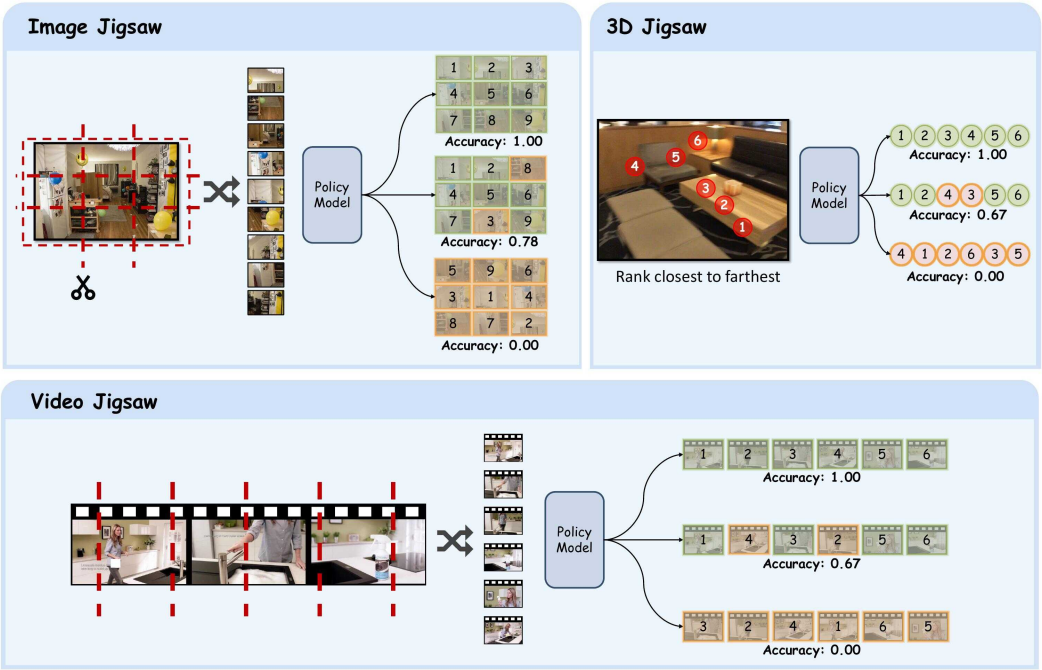

Figure 2: **Illustration of the Visual Jigsaw tasks. Left**: Image Jigsaw shuffles 3×3 spatial patches and asks the model to recover the original grid order. **Middle**: Video Jigsaw temporally reorders short video clips and asks the model to predict the chronological sequence. **Right**: 3D Jigsaw randomly permutes RGB-D frames ordered by scene depth and asks the model to reconstruct the front-to-back ordering.

a boxed answer. The ground-truth label is the inverse permutation $\pi^* = \sigma^{-1}$, which is directly computed from the shuffling process—no external annotations required.

3.2 Visual Jigsaw Instantiations

Image Jigsaw. For image data, we tile each image from the COCO 2017 training split (Lin et al., 2014) into a 3×3 grid of non-overlapping patches, yielding $n = 9$ patches per image. The patches are shuffled uniformly at random, and the model is presented with all nine patches in shuffled order and asked to output a permutation of indices 1–9 indicating the correct spatial arrangement (*i.e.*, which shuffled patch belongs at each grid position). This task demands fine-grained local visual understanding: the model must recognize object boundaries, texture continuity, and compositional cues that span patch boundaries to correctly reassemble the image.

Video Jigsaw. For video data, we sample a fixed number of short video clips from LLaVA-Video (Zhang et al., 2024b). Each clip is a temporally contiguous segment of the video, and clips are shuffled to form the input. The model must predict the chronological order of the n clips. This instantiation requires temporal understanding: recognizing the progression of events, changes in object state, or causal relationships that indicate temporal order. We train with $n = 8$ clips and evaluate at 16, 32, and 64 input frames.

3D Jigsaw. For 3D spatial understanding, we use RGB images from ScanNet indoor scenes (Dai et al., 2017). Frames from the same scene are selected and ordered by their estimated scene depth (distance from camera to scene center), from closest to farthest. The frames are shuffled and the model must recover the depth ordering—a task that requires understanding of 3D spatial relationships, perspective foreshortening, and scene geometry as observed in 2D images. This instantiation connects naturally to depth comparison and monocular depth perception benchmarks.

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3.3 GRADED REWARD DESIGN

A key challenge in jigsaw-style permutation prediction is designing a reward that is both informative and resistant to hacking. We propose a graded reward function $R(\hat{\pi}, \pi^*)$ that decomposes into three components:

(1) Format reward r_{fmt}: We require the response to follow the chain-of-thought format `<think>...\</think>` followed by a `\boxed{...}` answer. If the response does not match this format, the total reward is 0. This enforces explicit reasoning and prevents degenerate outputs. The format reward contributes $r_{\text{fmt}} = 0.2$ when satisfied.

(2) Accuracy reward r_{acc}: If the predicted permutation exactly matches the ground truth ($\hat{\pi} = \pi^*$), we assign a full accuracy reward of $r_{\text{acc}} = 1.0$. For partially correct predictions with a valid permutation (no repeated indices), we compute a discounted partial credit:

$$r_{\text{acc}} = \gamma \cdot \frac{|\{i : \hat{\pi}_i = \pi^*_i\}|}{n}, \tag{1}$$

where $\gamma = 0.2$ is the partial credit discount factor and this partial reward only applies for harder jigsaws with $n > 4$ (*i.e.*, the 3×3 image grid). This gradient provides a learning signal for near-correct solutions without incentivizing trivially easy sub-orderings.

(3) Repetition penalty r_{rep}: To prevent reward hacking via repetitive outputs, we apply a penalty of -0.5 if excessive repetition of long lines is detected in the generated response.

The total reward is:

$$R(\hat{\pi}, \pi^*) = r_{\text{fmt}} + r_{\text{acc}} + r_{\text{rep}}, \tag{2}$$

with the constraint that $R = 0$ when the format check fails. This design ensures that the model cannot obtain positive rewards through malformed or degenerate outputs.

3.4 TRAINING WITH GRPO

We train all Visual Jigsaw variants using Group Relative Policy Optimization (GRPO) (Shao et al., 2024), implemented via the verl framework (Sheng et al., 2024). GRPO samples $G = 16$ responses per prompt, normalizes rewards within the group to compute advantages, and optimizes the policy without a learned value function. We initialize from Qwen2.5-VL-7B-Instruct (Bai et al., 2025) and train for 32 epochs with a learning rate of 1×10^{-6}, batch size of 256 (image) or 128 (video and 3D), and a maximum response length of 2048 tokens. No KL divergence penalty is applied in the reward; the reference model is used only for log-probability computations during rollout. Training uses 8 A100 GPUs per node.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Base model. We use Qwen2.5-VL-7B-Instruct (Bai et al., 2025) as our base model for all main experiments. We also evaluate compatibility with the reasoning model ThinkLite-VL (Wang et al., 2025e).

Training data. Image Jigsaw uses images from COCO 2017 (Lin et al., 2014); Video Jigsaw uses videos from LLaVA-Video (Zhang et al., 2024b); 3D Jigsaw uses RGB-D frames from ScanNet (Dai et al., 2017).

Baselines. We compare against (i) Qwen2.5-VL-7B-Instruct (base), (ii) Image Jigsaw with SFT instead of RL, (iii) reasoning-oriented RL baselines: ThinkLite-VL (Wang et al., 2025e), VL-Cogito (Yuan et al., 2025a), and LLaVA-Critic-R1 (Wang et al., 2025d) on image benchmarks, and (iv) Video-R1 (Feng et al., 2025) on video benchmarks.

Evaluation. We evaluate on a comprehensive battery of benchmarks spanning fine-grained perception, spatial understanding, compositional understanding, temporal video reasoning, and 3D spatial perception, as detailed below.

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4.2 Image Perception Results

Figure 1 and Table 1 show results on image benchmarks. Image Jigsaw with RL consistently improves over the Qwen2.5-VL-7B baseline across all 13 evaluated benchmarks. Gains are especially pronounced on fine-grained perception tasks: MMVP (Tong et al., 2024) improves by +6.00 (54.66→60.66), MMStar fine-grained (Chen et al., 2024) by +6.06 (59.75→65.81), and HR-Bench-8K (Wang et al., 2025c) by +3.75 (67.38→71.13). Spatial understanding also benefits, with DA-2K (Yang et al., 2024) improving by +5.90 (54.45→60.35). Grounding (LISA (Lai et al., 2024): +2.65) and open-vocabulary detection (OVD-Eval (Yao et al., 2024): +1.42) also show positive transfer.

RL vs. SFT. A key finding is that RL significantly outperforms SFT. The SFT variant improves MMVP and HR-Bench but critically *degrades* LISA-Grounding (71.89→66.59, a drop of -5.30), while RL improves it (+2.65). This confirms prior observations (Chen et al., 2025) that SFT on perception tasks risks forgetting and degrading other capabilities, while RL provides more stable, broadly positive transfer.

Comparison to reasoning-oriented baselines. Image Jigsaw outperforms ThinkLite-VL, VL-Cogito, and LLaVA-Critic-R1—all reasoning-centric RL models—on the majority of perception benchmarks (MMVP, MMStar fine-grained, LISA-Grounding, VSR, DA-2K), despite those models being specifically designed for RL post-training. This demonstrates that vision-centric RL provides superior gains on perception tasks compared to reasoning-oriented RL.

Table 1: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
Model	**MMVP**	**MMStar (fine-grained)**	**MMBench**	**HR-Bench-8K**	**V***	**MME-RealWorld**	**LISA-Grounding**	**OVD-Eval**	**VSR**	**OmniSpatial**	**DA-2K**	**Winoground**	**SugarCrepe++**
	test	fine	en_dev	test	test	lite	test	test	test	test	val	g-acc	test
ThinkLite-VL	55.33	59.95	84.19	68.12	76.96	46.17	73.70	35.78	78.09	42.60	58.46	35.25	61.49
VL-Cogito	55.33	56.64	82.98	69.62	79.58	47.63	72.26	35.78	79.82	44.29	56.43	38.25	63.59
LLaVA-Critic-R1	53.33	57.80	83.16	67.50	78.01	45.18	68.52	35.28	78.50	42.73	53.82	34.75	61.93
Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	80.68	43.55	61.46	38.75	62.03
Image Jigsaw	60.66	65.81	84.45	71.13	80.63	45.96	74.54	36.49	80.36	44.49	60.35	39.00	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

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4.3 Video Perception Results

Table 2 shows results on 13 video benchmarks at three frame counts (16, 32, 64). Video Jigsaw consistently outperforms both the Qwen2.5-VL-7B baseline and Video-R1 (Feng et al., 2025) across all frame settings. At 64 frames, gains include: AoTBench (Xue et al., 2025) +5.23 (52.41→57.64), Vinoground (Zhang et al., 2024a) +3.40 (21.80→25.20), and CVBench (Zhu et al., 2025) +3.00 (51.50→54.50). Temporal benchmarks such as TOMATO (Shangguan et al., 2024) (+1.95) and FAVOR-Bench (Tu et al., 2025) (+1.41) also benefit, demonstrating improved temporal understanding.

Notably, performance scales consistently with the number of input frames: Video Jigsaw at 64 frames outperforms its 16-frame counterpart on all benchmarks, suggesting that the model leverages richer temporal context when available. Compared to Video-R1, which applies reasoning-oriented RL to video, Video Jigsaw achieves superior performance on temporal understanding benchmarks (AoTBench, Vinoground, TOMATO) while remaining competitive or superior on general video QA tasks (VideoMME, TempCompass).

Table 2: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	AoTBench	Vinoground	TOMATO	FAVOR-Bench	TUNA-Bench	VideoMME	TempCompass	TVBench	MotionBench	LVBench	VSI-Bench	Video-TT	CVBench
		vqa	group	test	test	test	wo subs	mc	test	val	test	test	mcq	test
Video-R1	16	45.06	9.40	27.29	49.47	53.00	56.62	70.19	51.80	55.82	34.53	34.34	42.95	47.50
Video-R1	32	47.53	10.20	27.29	49.90	54.26	59.88	71.77	53.54	56.12	38.61	35.11	42.63	48.10
Video-R1	64	48.68	10.60	27.36	50.51	54.33	60.85	72.59	53.43	56.09	38.80	36.61	42.74	48.69
Qwen2.5-VL-7B	16	45.52	12.60	25.87	48.54	53.14	57.44	71.77	49.94	55.56	33.51	32.79	38.39	47.70
Qwen2.5-VL-7B	32	49.48	18.20	26.34	49.34	54.88	60.70	72.59	51.96	56.47	39.19	35.34	41.57	49.60
Qwen2.5-VL-7B	64	52.41	21.80	26.35	50.86	55.79	63.44	72.84	53.74	56.29	40.35	37.74	42.25	51.50
Video Jigsaw	16	51.67	15.20	27.56	49.69	55.10	58.07	73.10	51.33	56.87	36.41	35.39	40.19	49.80
(Gain)		+6.15	+2.60	+1.69	+1.15	+1.96	+0.63	+1.33	+1.39	+1.31	+2.90	+2.60	+1.80	+2.10
Video Jigsaw	32	55.00	21.40	28.03	50.56	56.49	62.37	73.60	53.31	57.99	39.70	38.47	43.27	51.60
(Gain)		+5.52	+3.20	+1.69	+1.22	+1.61	+1.67	+1.01	+1.35	+1.52	+0.51	+3.13	+1.70	+2.00
Video Jigsaw	64	57.64	25.20	28.30	52.27	56.63	64.74	73.60	54.18	57.91	41.83	40.40	44.11	54.50
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

4.4 3D Spatial Perception Results

Table 3 shows results on 3D benchmarks. 3D Jigsaw yields the most dramatic improvements of all three variants. On the directly targeted depth-comparison benchmark DA-2K (Yang et al., 2024), performance jumps from 54.45 to 71.56 (+17.11). SAT-Real (Ray et al., 2024) improves from 48.66 to 64.00 (+15.34). Improvements generalize beyond depth comparison to broader 3D benchmarks: OmniSpatial (Jia et al., 2025) +3.33, ViewSpatial (Li et al., 2025) +2.10, VSI-Bench (Yang et al., 2025) +2.90, All-Angles (Yeh et al., 2025) +1.50, and SPARBench (Zhang et al., 2025) +2.56. These results demonstrate that training the model to reason about depth ordering from 2D images—a purely visual task—substantially enhances 3D spatial understanding across diverse input types (single-view, multi-view, and egocentric video).

Table 3: **Evaluation results on 3D benchmarks.** 3D Jigsaw consistently enhances performance on both directly related depth comparison tasks (DA-2K) and broader 3D perception tasks spanning single-view, multi-view, and egocentric video inputs.

Model	SAT-Real	3DSRBench	ViewSpatial	All-Angles	OmniSpatial	VSI-Bench	SPARBench	DA-2K
	test	test	test	test	test	test	tiny	test
Qwen2.5-VL-7B	48.66	57.42	36.52	47.56	42.66	37.74	35.75	54.45
3D Jigsaw	64.00	58.13	38.62	49.06	45.99	40.64	38.31	71.56
(Gain)	+15.34	+0.71	+2.10	+1.50	+3.33	+2.90	+2.56	+17.11

4.5 Compatibility with Reasoning Models

A potential concern is whether vision-centric RL post-training might interfere with existing reasoning capabilities. We address this by applying Image Jigsaw to ThinkLite-VL (Wang et al., 2025e), a reasoning-oriented MLLM trained with RL for mathematical reasoning. Table 4 shows that Image Jigsaw improves the vision-centric average score (59.69→61.60) while maintaining or improving reasoning performance: MathVista remains comparable (75.20→75.10), MathVision improves (30.92→35.20), and EMMA improves (26.75→29.00). These results confirm that Visual Jigsaw *complements* rather than conflicts with reasoning-oriented training.

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Table 4: Performance of Visual Jigsaw on a reasoning-oriented MLLM (ThinkLite-VL), showing improved visual perception while preserving reasoning ability.

Model	Vision-Centric Avg	MathVista testmini	MathVision testmini	MathVerse testmini	MMMU val	EMMA mini
ThinkLite-VL	59.69	75.20	30.92	50.76	55.11	26.75
+ Image Jigsaw	61.60	75.10	35.20	50.50	54.22	29.00

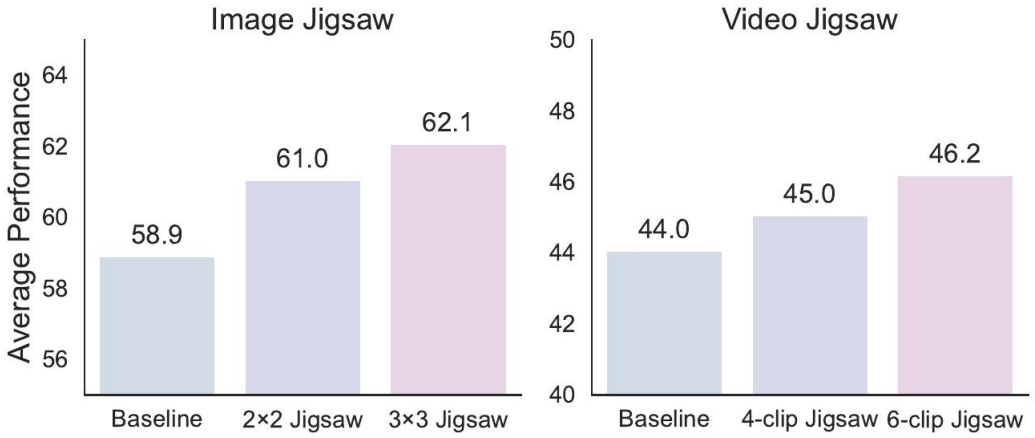

Figure 3: **Effect of jigsaw difficulty.** Performance on image (left) and video (right) benchmarks as a function of jigsaw size (number of pieces/clips). Moderate difficulty provides the best downstream performance; too-easy jigsaws underfit while too-hard jigsaws may be over-challenging for stable RL training.

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4.6 Effect of Jigsaw Difficulty

We ablate the difficulty of the jigsaw task by varying the number of pieces. Figure 3 shows results on representative image and video benchmarks. A moderate number of pieces provides the best downstream performance: too few pieces (easy jigsaws) provide insufficient learning signal, while excessively many pieces create overly hard tasks that destabilize RL training. The 3×3 grid (9 pieces) and 8 temporal clips represent the operating points used in our main experiments.

4.7 Discussion

Why does jigsaw solving improve downstream perception? The jigsaw task forces the model to attend to local visual details, spatial relationships, temporal dynamics, and depth cues that are often ignored during standard instruction tuning. By making permutation recovery the explicit objective, the model learns more discriminative visual representations transferable to diverse downstream perception tasks.

Why RL over SFT? SFT on permutation labels directly supervises the output sequence, which can lead to forgetting of other capabilities and overfit to the jigsaw format. RL with a graded reward allows the model to explore diverse reasoning strategies and receive informative feedback even for partially correct solutions, leading to more robust and general visual representations.

Plug-and-play applicability. Because Visual Jigsaw requires only text outputs (permutation indices) and standard visual inputs, it is immediately applicable to any text-output MLLM without modification. The training data is derived from freely available datasets (COCO, LLaVA-Video, ScanNet), further reducing the barrier to adoption.

5 CONCLUSION

We presented **Visual Jigsaw**, a self-supervised, vision-centric post-training framework that recasts visual understanding as a verifiable permutation ordering task. By instantiating this framework across images (spatial patch ordering), videos (temporal clip ordering), and 3D data (depth-ordering of RGB-D frames), and training with GRPO using a carefully designed graded reward, we achieve substantial and consistent improvements over the Qwen2.5-VL-7B baseline across a diverse suite of visual benchmarks. Visual Jigsaw outperforms SFT, surpasses reasoning-centric RL baselines on perception tasks, and complements reasoning-oriented models without degrading their mathematical skills. Its simplicity, scalability, and plug-and-play nature make it a broadly applicable recipe for strengthening intrinsic visual understanding in MLLMs.

Limitations. While Visual Jigsaw significantly improves visual perception, the permutation task itself is a proxy that may not capture all dimensions of visual understanding (e.g., semantic grounding or visual commonsense). Future work may explore combining Visual Jigsaw with complementary training signals. Additionally, scaling to larger models and more difficult jigsaw configurations remains an open direction.

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